# Where Quantized Language Models Break: Task-Specific Accuracy Cliffs Lie Below Four Bits, and KL Divergence Marks Them Without Warning

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## Abstract

Weight quantization is the default way to deploy open language models, yet bit widths are usually chosen from perplexity or a single multiple-choice benchmark. We ask which task types lose accuracy first as precision drops, whether the loss is gradual or a cliff, whether smaller models break at higher precision, and whether KL divergence from the full-precision model warns of the cliff in advance. We quantize seven instruction-tuned models (0.5B–8B; QWEN2.5, LLAMA-3, PHI-3-MINI) with weight-only round-to-nearest at 8, 6, 5, 4, 3 and 2 bits (group size 128) and score them on six task types: arithmetic, factual QA, reading comprehension, code, instruction following and creative writing. The cliff lies at or below 4→3 bits for every task, not between INT8 and INT4. Eight and six bits are essentially lossless (3 significant drops in 72 paired tests). At 4 bits the loss is modest but task-dependent: code is the most sensitive task (mean retained score 0.84) and instruction following the least (0.97). Larger models are more robust (Spearman $\rho = -0.53$ between size and critical precision), but not monotonically. Task-specific KL separates broken from healthy configurations with AUC 0.97 at a threshold near 0.21 nats. However, it is a concurrent signal rather than a leading one: one level before failure it exceeds 0.3 nats in only 6 of 42 cases. Aggregate accuracy also hides churn: 7–15% of correct answers flip at 5 bits, and over 90% of 4-bit errors are fluent and confident. Deployers of INT4 models should therefore validate each task directly rather than rely on perplexity or a single KL threshold.

## 1 Introduction

Most open language models are deployed with quantized weights. Storing weights in 4 or 8 bits instead of 16 cuts memory by 2–4× and makes a 7B model fit on a laptop GPU. The cost of this compression is usually summarized by one number: a perplexity ratio on WIKITEXT-2, or accuracy on one multiple-choice benchmark Frantar et al. [2023], Lin et al. [2024], Dettmers and Zettlemoyer [2023]. People deploy these models for very different jobs, however, such as writing code, solving word problems, answering factual questions or drafting stories. If quantization damages these tasks unevenly, a single aggregate number can approve a bit width that silently breaks the one task a user cares about.

**What do practitioners believe about quantization damage?** A common view is that INT8 is safe, INT4 is "mostly fine", and that tasks needing precise reasoning or memorized facts break first, possibly at a sharp cliff between INT8 and INT4. A second common view is that small models are more fragile than large ones. A third is that a distribution-level signal, such as the KL divergence between the quantized and full-precision next-token distributions, can flag trouble before task accuracy visibly drops. Each view is plausible, but none has been tested with a sweep fine enough to *locate* a cliff across many task types.

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**What is missing?** Quantization papers such as GPTQ Frantar et al. [2023] and AWQ Lin et al. [2024] report perplexity and a few zero-shot multiple-choice tasks, mostly at 3 and 4 bits. Later evaluation studies Li et al. [2024], Jin et al. [2024], Marchisio et al. [2024] show that degradation is uneven across tasks, and Dutta et al. [2024] argue that accuracy hides "flips" and should be complemented by divergence metrics. To our knowledge, however, no study combines (i) a bit-width sweep fine enough to locate the cliff, (ii) generative tasks such as code, instruction following and creative writing alongside QA, and (iii) an explicit test of whether KL divergence *leads* the accuracy cliff, with a numeric threshold.

**Our approach.** We run a controlled sweep: seven instruction-tuned models from three families (QWEN2.5 0.5B/1.5B/3B/7B, LLAMA-3 1B/8B, PHI-3-MINI 3.8B), seven weight precisions (bf16, 8, 6, 5, 4, 3, 2 bits; round-to-nearest, group size 128), and six task types with 1,614 prompts in total. For every configuration we also measure the task-specific KL divergence from the bf16 model on the same task distribution. We pre-register four hypotheses. Arithmetic and factual QA should drop by more than 15% between INT8 and INT4 (H1). Creative writing and reading comprehension should stay above 85% at INT4 (H2). Smaller models should cliff at higher precision (H3). KL should cross about 0.3 nats at or before the cliff (H4). We complement the sweep with official AWQ INT4 checkpoints and a per-channel quantization ablation.

**What do we find?** Most of the hypotheses fail in their specific form, and the ways they fail are informative. The cliff for every task lies at or below 4→3 bits: the largest drop is at 4→3 bits in 28 of 30 cells for models up to 3.8B, and at 3→2 bits in all 12 cells for the 7–8B models (figure 1). Eight and six bits are essentially lossless, with 3 significant drops in 72 paired model–task tests. At 4 bits, *code* is the most sensitive task, not arithmetic: mean retained score is 0.84 for HUMANEVAL versus 0.90 for GSM8K and 0.97 for IFEVAL. Size helps (Spearman $\rho = -0.53$, $p = 0.0003$), but QWEN2.5 3B is the least robust QWEN2.5 model at 3 bits. Task-specific KL detects broken configurations with AUC 0.97 and an optimal threshold of 0.21 nats (95% CI 0.12–0.34), consistent with the hypothesized 0.3. It does not warn early, though: at the last healthy precision before a failure, KL exceeds 0.3 nats in only 6 of 42 cases. Finally, aggregate accuracy hides heavy answer churn, and more than 90% of the answers lost at 4 bits are fluent and confident.

Our contributions are:

- We map task-specific accuracy cliffs for 7 models $\times$ 7 precisions $\times$ 6 task types and show that the cliff lies between 4 and 3 bits for $\leq$3.8B models and between 3 and 2 bits for 7–8B models, not between INT8 and INT4.
- We show that code generation is the most INT4-sensitive task and instruction following the least. We explain the ranking by whether correctness requires every token to be exact.
- We test KL divergence as an early-warning signal and find that it is an excellent *concurrent* detector (AUC 0.97, threshold $\approx$0.2–0.3 nats) but not a *leading* one, and that its useful threshold varies by about $100\times$ across tasks.
- We show that a fixed "INT4" label is not a fixed accuracy level. Removing the group-wise scales costs 30 points of retained score at 4 bits, while AWQ recovers most of the round-to-nearest loss. Silent, fluent errors and answer churn are hidden by aggregate accuracy.

Section 2 reviews related work, section 3 describes the setup, section 4 presents the results for each hypothesis, and section 5 discusses implications and limitations.

## 2 Related Work

**Post-training weight quantization.** Dettmers et al. [2022] showed that 8-bit inference preserves quality at scale once outlier features are handled. GPTQ Frantar et al. [2023] uses approximate second-order information to quantize weights to 3–4 bits, and AWQ Lin et al. [2024] protects salient channels identified from activation statistics. SmoothQuant Xiao et al. [2023] moves activation outliers into the weights to enable joint weight–activation quantization. Dettmers and Zettlemoyer [2023] study the trade-off between bit width and parameter count and argue that 4 bits is close to optimal for a fixed memory budget. These papers evaluate mainly with perplexity and a few zero-shot multiple-choice tasks. We instead use the simplest quantizer, round-to-nearest (RTN), as a pessimistic and calibration-free lower bound, and sweep it over six bit widths to locate cliffs. We use official AWQ checkpoints only to check that our INT4 conclusions transfer to a production method.

**Evaluating quantized models beyond perplexity.** Several studies show that quantization damage is uneven. Li et al. [2024] evaluate quantized models on basic NLP, emergent abilities, trustworthiness and dialogue. Jin et al. [2024] find that 4-bit models keep most of their performance while lower precisions degrade sharply. Marchisio et al. [2024] find that automatic metrics understate the damage in non-Latin-script languages and that human judges notice drops that benchmarks miss. These studies mostly compare a few standard settings (e.g., W8, W4, W3). Our sweep adds 6- and 5-bit points so that the cliff can be located, and it places generative tasks (code, instruction following, creative writing) under one protocol with QA tasks.

**Distributional metrics for compressed models.** Dutta et al. [2024] show that compressed models can match baseline accuracy while changing many individual answers ("flips"), and they recommend KL divergence and flip rates as complementary metrics. We confirm the churn they describe on generative tasks. We also go one step further and test whether task-specific KL is a *leading* indicator of the accuracy cliff, i.e., whether it crosses a fixed threshold one precision level before accuracy fails. We find that it does not.

**Benchmarks.** Our task suite uses GSM8K Cobbe et al. [2021], TRIVIAQA Joshi et al. [2017], SQUAD Rajpurkar et al. [2016], HUMANEVAL Chen et al. [2021], IFEVAL Zhou et al. [2023] and WRITINGPROMPTS Fan et al. [2018], together with WIKITEXT-2 Merity et al. [2016] perplexity. For creative writing we use an LLM judge, following Zheng et al. [2023].

# 3 Experimental Setup

## 3.1 Problem formulation

Let $s_m^t(b)$ be the score of model $m$ on task $t$ when its weights are quantized to $b$ bits, with $b = 16$ denoting the bf16 reference. We define the **retained fraction**

$$\mathrm{ret}_m^t(b) = s_m^t(b)/s_m^t(16). \tag{1}$$

Following our pre-registered plan, a **cliff** is a drop in ret of more than 0.15 between two adjacent tested precisions. The **critical precision** ${b^\star}_{m,t}$ is the highest tested $b$ with $\mathrm{ret}_m^t(b) < 0.85$; a configuration with $\mathrm{ret} < 0.85$ is called **broken**. For each configuration we also compute the token-level divergence

$$D_{\mathrm{KL}}\big(p_{16} \,\|\, p_b\big) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \sum_{v} p_{16}(v \mid x_{<i}) \log \frac{p_{16}(v \mid x_{<i})}{p_b(v \mid x_{<i})}, \tag{2}$$

averaged over the response tokens $\mathcal{T}$ of task-specific prompts. Here $p_{16}$ and $p_b$ are the bf16 and quantized next-token distributions under the same prefix.

## 3.2 Models and quantization

**Models.** We use seven instruction-tuned bf16 checkpoints: QWEN2.5-Instruct 0.5B, 1.5B, 3B and 7B Qwen Team [2024], LLAMA-3.2-1B-INSTRUCT and LLAMA-3.1-8B-INSTRUCT Grattafiori et al. [2024], and PHI-3-MINI-4k-Instruct (3.8B) Abdin et al. [2024]. As a production reference, we add the official QWEN2.5-3B/7B-Instruct-AWQ INT4 checkpoints (group size 128). Quantization requires weight access, so all models run locally and no API models are involved.

**Quantizer.** We apply weight-only, asymmetric min–max round-to-nearest quantization with group size 128 along the input dimension. It covers every linear layer inside the transformer blocks (attention q/k/v/o and MLP gate/up/down), while embeddings and the LM head stay in bf16, as in the GPTQ and AWQ evaluations. We use fake quantization: weights are rounded to the $2^b$-level grid of their group and de-quantized back to bf16, which is numerically what an INT-$b$ group-128 kernel computes. Unit tests check that every group has at most $2^b$ distinct values. We choose RTN for two reasons. It needs no calibration data, which removes the choice of calibration set as a confound. It is also a pessimistic lower bound for better quantizers, which the AWQ checkpoints let us quantify. For the granularity ablation (section 4.5) we also run per-output-channel RTN without grouping, at 8/6/5/4/3 bits, for QWEN2.5-1.5B and LLAMA-3-1B.

**Precision levels.** Every model is run at bf16, 8, 6, 5, 4, 3 and 2 bits, except PHI-3-MINI, which was run only at bf16, 4 and 3 bits because of GPU time constraints. All analyses use only the levels actually run, and cliffs are computed between adjacent *tested* levels.

Table 1: Task suite. Each task type uses one dataset, a fixed random subset, and a task-appropriate metric. `max_tokens` is the generation limit.

| Task type | Dataset | Prompt | $n$ | Metric | max_tokens |
|---|---|---|---|---|---|
| Arithmetic reasoning | GSM8K test | zero-shot CoT, "The answer is N" | 250 | exact numeric match | 512 |
| Factual QA | TRIVIAQA rc.nocontext val | closed-book short answer | 400 | normalized alias match | 32 |
| Reading comprehension | SQUAD v1.1 val | passage + question, extract span | 400 | token F1 | 32 |
| Code generation | HUMANEVAL (all) | complete the function | 164 | pass@1 (sandboxed) | 512 |
| Instruction following | IFEVAL (300 of 541) | raw prompt | 300 | prompt-level strict acc. | 768 |
| Creative writing | WRITINGPROMPTS test | ∼200-word story | 100 | LLM judge (1–10, rescaled) | 400 |

### 3.3 Tasks and metrics

Table 1 lists the six task types. All subsets are fixed with seed 42 (1,614 prompts in total). Decoding is greedy (temperature 0) with each model's chat template, so every configuration has a single deterministic output.

**Creative-writing judge.** Each story is scored by Qwen3-8B-AWQ Qwen Team [2025] (thinking disabled, greedy decoding). The judge rates fluency, coherence, relevance and creativity on a 1–10 scale, and the score is their mean divided by 10. The judge is blind to model and precision, and it produced no parse failures.

### 3.4 Divergence metrics

**Task-specific KL.** For each model and task, we take the bf16 greedy responses to the first 60 items (truncated to 256 tokens) and teacher-force them through every quantized version. The KL is computed from the bf16 top-128 log-probabilities plus one lumped tail bucket. This gives a lower bound on the exact KL, which we verified to be within about 1% of it (e.g., 0.3624 vs. 0.3661 nats). We also record top-1 agreement with bf16.

**Perplexity.** As the standard aggregate proxy, we compute WIKITEXT-2 test perplexity on 32 windows of 1,024 tokens and report it as a ratio to bf16.

### 3.5 Statistical analysis

We report 95% bootstrap confidence intervals (1,000 resamples over items) for every score. To test each quantized level against bf16, we use paired McNemar tests for binary metrics and paired Wilcoxon tests for F1 and judge scores. We correct the 160 resulting tests with the Benjamini–Hochberg procedure Benjamini and Hochberg [1995] at $q < 0.05$. A Friedman test across models checks whether tasks differ in sensitivity, and a Spearman correlation relates log parameter count to critical precision. For H4, we compute the ROC-AUC of KL for classifying broken configurations, the Youden-optimal threshold with a bootstrap CI, and an out-of-sample check on the per-channel runs. We also run a lead-time analysis that compares KL at the last healthy precision with KL at the failing one.

**Implementation.** Generation uses vLLM Kwon et al. [2023]. Quantization is applied in place inside the vLLM worker, starting from a CPU copy of the original weights, and prefix caching is disabled. KL and perplexity use HF transformers with the same quantization code. Experiments ran on shared RTX A6000 GPUs and produced about 9.8M generated tokens, with no API cost.

## 4 Results

We organize the results around the four hypotheses. Section 4.1 locates the cliff and ranks tasks (H1, H2), section 4.2 examines what quantization errors look like, section 4.3 tests the effect of model size (H3), section 4.4 tests KL as an early warning (H4), and section 4.5 ablates quantization granularity. Table 7 in the appendix lists the bf16 reference scores with confidence intervals.

### 4.1 Where is the cliff? (H1, H2)

**8 to 5 bits are essentially lossless.** Table 2 shows that mean retained scores stay between 0.95 and 1.02 from 8 to 5 bits for every task. At 8 and 6 bits only 3 of 72 paired model–task tests show a

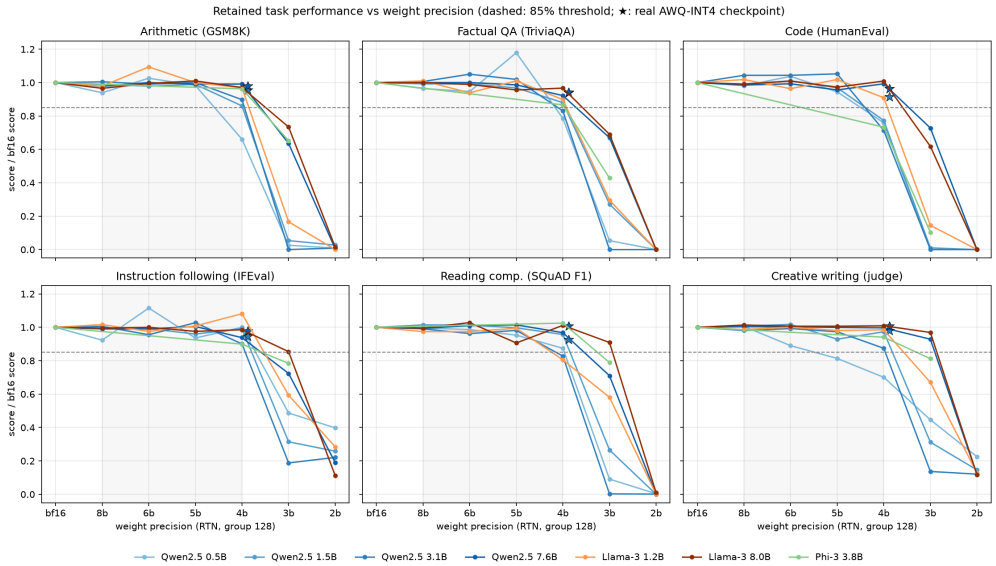



Figure 1: Retained score (quantized / bf16) as weight precision drops, for each task type. Lines show RTN group-128 per model, and stars mark official AWQ INT4 checkpoints. The dashed line marks 0.85 retained, and the grey band spans INT8→INT4. All curves are flat from 8 to 5 bits. Most curves drop sharply between 4 and 3 bits, and the 7–8B models (dark lines) drop mainly between 3 and 2 bits. AWQ stars sit close to 1.0 at 4 bits even when RTN INT4 does not.

Table 2: Mean retained fraction across models (RTN, group 128), with the number of models showing a significant drop vs. bf16 (paired test, BH-corrected $q < 0.05$) in small type. Six models are tested at 8/6/5/2 bits and seven at 4/3 bits. The most robust task at each of 4, 3 and 2 bits is in **bold**, and the most fragile is underlined. †IFEVAL has a non-zero floor, because degenerate output trivially satisfies some constraints.

| Task | 8 bits | 6 bits | 5 bits | 4 bits | 3 bits | 2 bits |
|---|---|---|---|---|---|---|
| Arithmetic (GSM8K) | 0.98 0/6 | 1.01 0/6 | 1.00 0/6 | 0.90 3/7 | 0.32 7/7 | 0.01 6/6 |
| Factual QA (TRIVIAQA) | 1.00 0/6 | 0.99 0/6 | 1.02 1/6 | 0.88 3/7 | 0.34 7/7 | 0.00 6/6 |
| Code (HUMANEVAL) | 1.00 0/6 | 1.01 0/6 | 0.99 0/6 | 0.84 3/7 | 0.23 7/7 | 0.00 6/6 |
| Instruction following (IFEVAL) | 0.99 0/6 | 1.00 0/6 | 0.99 0/6 | **0.97** 0/7 | 0.56 7/7 | **0.24**† 6/6 |
| Reading comp. (SQUAD F1) | 0.99 1/6 | 1.00 1/6 | 0.97 1/6 | 0.92 5/7 | 0.48 7/7 | 0.00 6/6 |
| Creative writing (judge) | 1.00 0/6 | 0.99 1/6 | 0.95 2/6 | 0.93 3/7 | **0.61** 7/7 | 0.14 6/6 |

significant drop (one small SQUAD dip at 8 bits; one SQUAD and one creative-writing drop at 6 bits). At 5 bits, 4 of 36 tests are significant. INT8 is statistically indistinguishable from bf16 on every task except one small SQUAD dip.

**The steepest drop is at 4→3 bits, not INT8→INT4.** For models up to 3.8B, the largest single-step drop occurs at 4→3 bits in 28 of 30 model–task cells; the exceptions are LLAMA-3-1B on reading and creative writing. For the 7–8B models it occurs at 3→2 bits in all 12 cells. The pre-registered INT8→INT4 cliff (>0.15 retained drop, among the 6 models with both levels) is rare: it appears for 1 of 6 models on arithmetic (QWEN2.5-0.5B), 2 of 6 on factual QA, 3 of 6 on code, 2 of 6 on reading, 1 of 6 on creative writing and 0 of 6 on instruction following. **H1 is not supported as stated.**

**Code, not arithmetic, is most sensitive at INT4.** Arithmetic and factual QA are among the more sensitive tasks at 4 bits (mean retained 0.90 and 0.88, with significant drops in 3 of 7 models each). Code generation, however, is the most sensitive: mean retained is 0.84, and four models (QWEN2.5-0.5B/1.5B/3B and PHI-3-MINI) keep only 0.71–0.77 of their bf16 pass@1. The INT4 task ranking is not statistically significant (Friedman $p = 0.11$). At 3 bits the ranking is clear (Friedman $p = 10^{-4}$):

Table 3: Answer flips relative to bf16, pooled over models. *Lost* counts items correct in bf16 but wrong after quantization, and *gained* counts the reverse. *Fluent* is the share of lost items whose output terminated normally, i.e., without a repetition loop or hitting the token limit. Rows pool different model subsets per level (PHI-3-MINI only at 4/3 bits). Below 4 bits, losses far exceed gains, and the errors change from fluent to degenerate.

| | 8 bits | | | 5 bits | | | 4 bits | | | 3 bits | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Task** | Lost | Gained | Fluent | Lost | Gained | Fluent | Lost | Gained | Fluent | Lost | Gained | Fluent |
| Arithmetic | 63 (6%) | 41 | 0.86 | 112 (11%) | 108 | 0.93 | 211 (17%) | 105 | 0.90 | 827 (66%) | 34 | 0.30 |
| Factual QA | 17 (2%) | 16 | 1.00 | 80 (9%) | 75 | 1.00 | 214 (19%) | 95 | 1.00 | 698 (61%) | 46 | 0.68 |
| Code | 16 (3%) | 17 | 1.00 | 58 (10%) | 50 | 0.95 | 162 (24%) | 58 | 0.93 | 506 (74%) | 11 | 0.42 |
| Instruction following | 76 (8%) | 69 | 0.87 | 139 (15%) | 131 | 0.82 | 204 (19%) | 165 | 0.73 | 557 (53%) | 117 | 0.38 |
| Reading comp. | 29 (1%) | 17 | 0.90 | 131 (7%) | 88 | 0.93 | 309 (14%) | 141 | 0.92 | 1274 (57%) | 68 | 0.49 |

code collapses hardest (0.23), followed by arithmetic (0.32), factual QA (0.34) and reading (0.48), while instruction following (0.56) and creative writing (0.61) degrade least.

**Reading and creative writing are robust at INT4, except for the smallest model.** Reading comprehension keeps $\geq$0.85 of its bf16 F1 at INT4 for 5 of 7 models, with LLAMA-3-1B (0.81) and QWEN2.5-3B (0.83) just below. The SQUAD drops are small but consistent, so they reach significance more often than for any other task (5 of 7 models). Creative writing is robust at INT4 for all models from 1.2B up (0.87–1.01 retained). QWEN2.5-0.5B, however, degrades gradually from 6 bits (0.89, 0.81, 0.70 at 6, 5, 4 bits). This is the only configuration that falls below 0.85 at a precision above 4 bits. **H2 is partially supported.**

**A better quantizer recovers most of the INT4 loss.** The official AWQ INT4 checkpoints retain 0.91–0.98 on every task for QWEN2.5-3B and 0.94–1.01 for QWEN2.5-7B (table 8 in the appendix). For QWEN2.5-3B, AWQ raises code from 0.71 to 0.91, arithmetic from 0.90 to 0.95, and factual QA from 0.83 to 0.94. With a production quantizer, INT4 is close to lossless for models of 3B and larger.

### 4.2 What do quantization errors look like?

**Accuracy hides churn.** Table 3 shows that at 5 bits, where net accuracy is flat, 7–15% of the items solved in bf16 are answered wrongly, and a similar number of new items become correct. The quantized model has about the same average accuracy as the original, but it gives different answers to many individual items. Even at 8 bits, 6% of correct arithmetic answers and 8% of satisfied IFEVAL prompts flip.

**INT4 errors are silent; INT3 errors are loud.** At 4 bits, 90–100% of lost arithmetic, factual QA, code and reading items are fluent, confident wrong answers. Typical factual QA examples are: the founder of the World Cup, *Jules Rimet* $\rightarrow$ *Jules Renard*; the top album of 1985 alongside *Like A Virgin*, *Madonna* $\rightarrow$ *Michael Jackson*; the run of Star Trek: The Original Series, 3 $\rightarrow$ 4 years. Errors like these are hard to catch without labels. At 3 bits most failures are degenerate instead (only 30–68% fluent): repetition loops that hit the token limit, or truncated rambling. Such failures are easy to detect automatically.

### 4.3 Does size protect? (H3)

**Larger models are more robust, mostly at the extremes.** Critical precision correlates negatively with log parameter count (Spearman $\rho = -0.53$, $p = 0.0003$, $n = 42$ model–task cells; within QWEN2.5 alone, $\rho = -0.46$, $p = 0.02$). Table 4 shows that the effect is carried by the extremes. QWEN2.5-0.5B is the only model that already breaks at 4 bits on arithmetic, the only one with a critical precision of 4 bits on three tasks, and the only one that breaks at 5 bits (creative writing). The 7–8B models are the only ones that remain partly functional at 3 bits. QWEN2.5-7B keeps 0.64 on arithmetic, 0.73 on code and 0.67 on factual QA, and LLAMA-3-8B keeps 0.73, 0.62 and 0.69. LLAMA-3-8B also stays at or above 0.85 on instruction following, reading and creative writing, and QWEN2.5-7B on creative writing. Models of 3B and below fall to 0–0.3 on most tasks.

**QWEN2.5-3B breaks monotonicity.** QWEN2.5-3B is the most fragile QWEN2.5 model at 3 bits. It scores 0.00 retained on GSM8K, HUMANEVAL and TRIVIAQA, and its WIKITEXT-2 perplexity grows by 743$\times$, compared with 2.7$\times$ and 4.3$\times$ for the 1.5B and 0.5B models (table 9 in the ap-

Table 4: Critical precision $b^\star$: the highest tested bit width at which the model keeps less than 85% of its bf16 score; lower is more robust. Models are sorted by size. Cells at 4 bits or above are in **bold**. PHI-3-MINI was tested only at bf16/4/3 bits.

| Model | Params | Arith. | Fact. QA | Code | Instr. | Reading | Creative | Mean |
|---|---|---|---|---|---|---|---|---|
| QWEN2.5-0.5B | 0.49B | **4** | **4** | **4** | 3 | 3 | **5** | 3.8 |
| LLAMA-3-1B | 1.24B | 3 | 3 | 3 | 3 | **4** | 3 | 3.2 |
| QWEN2.5-1.5B | 1.54B | 3 | 3 | **4** | 3 | 3 | 3 | 3.2 |
| QWEN2.5-3B | 3.09B | 3 | **4** | **4** | 3 | **4** | 3 | 3.5 |
| PHI-3-MINI | 3.82B | 3 | 3 | **4** | 3 | 3 | 3 | 3.2 |
| QWEN2.5-7B | 7.62B | 3 | 3 | 3 | 3 | 3 | 2 | 2.8 |
| LLAMA-3-8B | 8.03B | 3 | 3 | 3 | 2 | 2 | 2 | 2.5 |

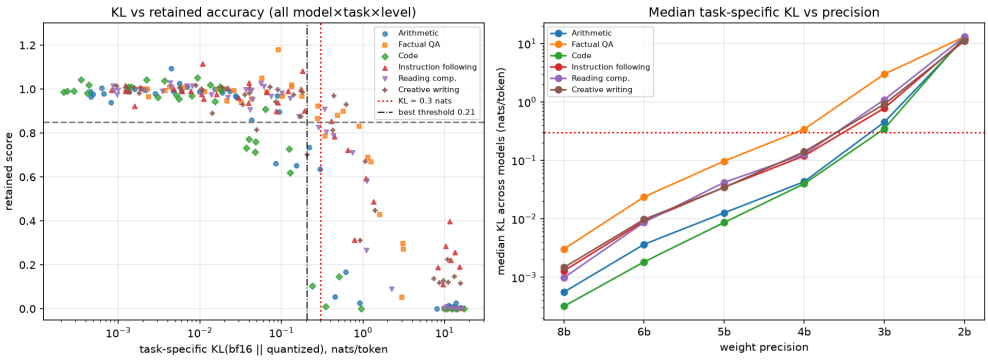



Figure 2: *(Left)*Task-specific KL(bf16 || quantized) vs. retained score for every model × task × level (log x-axis). The red dotted line marks the hypothesized 0.3 nats, and the dash-dotted line the fitted optimum of 0.21 nats. Broken configurations (below the dashed 0.85 line) mostly lie to the right of the threshold. The exception is a cluster of code failures (green diamonds) at KL ≈ 0.04–0.1. *(Right)*Median KL across models grows by roughly 5–10× per precision step. Factual QA has the highest KL at every level, and code the lowest.

pendix). At INT4 it also loses more on code and factual QA than the 1.5B model. One plausible cause is a small set of outlier weight channels that min–max RTN handles poorly. Consistent with this, AWQ, which protects salient channels, restores it to 0.91–0.98 retained at INT4. We did not verify this mechanism directly. **H3 is supported in direction but not monotonically**: the claim that smaller models cliff at higher precision holds clearly for 0.5B versus 7–8B, but not step by step within a family.

### 4.4 Does KL divergence predict the cliff? (H4)

**KL is an excellent concurrent detector.** Across 228 quantized model–task–level cells, 85 of them broken, KL correlates strongly with retained score (Spearman $-0.81$). Used as a classifier of broken configurations, KL reaches an AUC of 0.97, compared with 0.98 for the WIKITEXT-2 perplexity ratio and 0.94 for top-1 agreement. Per-task AUCs range from 0.96 to 1.00. The Youden-optimal threshold is **0.21 nats (bootstrap 95% CI 0.12–0.34)**, consistent with the hypothesized ≈0.3. At 0.3 nats, sensitivity is 0.84 and specificity 0.96. The result also holds out of sample: on the 60 per-channel cells, which were never used for fitting, KL reaches AUC 0.97, and the 0.3-nat rule has sensitivity 0.82 and specificity 0.92.

**KL does not warn early.** For each of the 42 model–task pairs that eventually fall below 0.85, we take the last healthy precision directly above the failing one. There, KL is at least 0.3 nats in only **6 of 42 cases (14%)**, with a median of 0.09 nats, and most of these 6 cases are factual QA, where healthy KL is already high. At the failing level itself, KL is at least 0.3 nats in 30 of 42 cases (71%), with a median of 0.70. KL rises by about 5–10× per precision step (figure 2, right), but it crosses 0.3 *together with* the accuracy drop, not before it.

Table 5: Median task-specific KL(bf16 || quantized) in nats per token across models. Values at or above the hypothesized 0.3-nat threshold are in **bold**. Factual QA crosses 0.3 nats at 4 bits even though most models remain healthy there, while code fails at 4 bits for four models with KL below 0.05.

| Task | 8 bits | 6 bits | 5 bits | 4 bits | 3 bits | 2 bits |
|---|---|---|---|---|---|---|
| Arithmetic | 0.001 | 0.004 | 0.013 | 0.043 | **0.45** | **12.2** |
| Factual QA | 0.003 | 0.024 | 0.097 | **0.34** | **3.00** | **13.1** |
| Code | 0.000 | 0.002 | 0.009 | 0.040 | **0.35** | **13.1** |
| Instruction following | 0.001 | 0.009 | 0.035 | 0.120 | **0.79** | **11.7** |
| Reading comp. | 0.001 | 0.009 | 0.042 | 0.128 | **1.10** | **13.1** |
| Creative writing | 0.001 | 0.010 | 0.034 | 0.141 | **0.92** | **11.0** |

Table 6: Mean retained fraction over all six tasks for QWEN2.5-1.5B and LLAMA-3-1B, with group-128 vs. per-output-channel RTN. Without group-wise scales the cliff moves up by a full bit: per-channel INT4 (0.62) sits between group-128 INT4 (0.92) and INT3 (0.31). Better (higher) value per row in **bold**.

| Granularity | 8 bits | 6 bits | 5 bits | 4 bits | 3 bits |
|---|---|---|---|---|---|
| Group 128 | 1.00 | **1.00** | 0.98 | **0.92** | **0.31** |
| Per-channel | 1.00 | 0.99 | 0.98 | 0.62 | 0.08 |

**The right threshold varies about $100\times$ across tasks.** HUMANEVAL breaks at very low KL: for the four models whose code cliffs at 4 bits, median KL at the failing level is only 0.04–0.05 nats, because one wrong token can fail the unit tests. TRIVIAQA tolerates KL of about 0.3–0.5 nats at healthy levels, because its answers are short and a few high-entropy answer tokens dominate the average (table 5). A single 0.3-nat rule therefore misses code failures and over-flags factual QA. The pooled threshold works mostly because KL at 3 and 2 bits is enormous. **H4 is partially supported**: KL is a strong concurrent signal with a pooled threshold of 0.2–0.3 nats, but it is not a leading indicator, and a pooled threshold is unsafe for code.

### 4.5 Ablation: quantization granularity

Table 6 shows that removing the group-wise scales moves the cliff up by a full bit. At 4 bits, per-channel RTN retains only 0.62 of bf16 performance on average, compared with 0.92 for group-128. Per-channel INT4 loses 33–57% on arithmetic, code and factual QA for both models, and QWEN2.5-1.5B's creative-writing score falls to 0.21. "INT4" therefore does not name a fixed accuracy level. The scale granularity and the quantizer (section 4.1, AWQ) matter as much as the bit count.

## 5 Discussion

**Why is the cliff below 4 bits?** Our data place the cliff between 4 and 3 bits for models up to 3.8B, which matches the GPTQ and AWQ papers: 4-bit group-128 is near-lossless there, and 3-bit RTN breaks down Frantar et al. [2023], Lin et al. [2024]. It contradicts the hypothesis that the cliff sits between INT8 and INT4. One explanation is that with 128-element groups, each group has its own scale and offset, so sixteen levels per group are enough to keep most weights close to their original values, while eight levels are not. The per-channel ablation supports this reading: with a single scale per output channel, the same 16 levels behave roughly like grouped 3.5 bits.

**Which tasks break first, and why?** The task ranking follows a simple rule: *tasks whose correctness depends on every token being exact degrade most*. Code passes its unit tests only if the whole function is right, and GSM8K needs an exact reasoning chain and final number. TRIVIAQA needs the retrieval of rare memorized facts, which are plausibly stored in small weight differences that rounding easily perturbs. Tasks with many acceptable outputs degrade least. IFEVAL constraints can be satisfied in many ways, stories are judged holistically, and SQUAD answers can be copied from the passage. This rule also explains why code, not arithmetic, tops the INT4 ranking: code outputs are longer, and any single error is fatal.

**What should practitioners do?** First, group-128 INT6 and INT8 are safe for all six task types, and INT5 is nearly so. Second, INT4 needs task-specific validation, especially for code and for models of 3B and below, and a calibrated quantizer such as AWQ recovers most of the RTN-INT4 loss. Third, RTN INT3 is unusable below 7B. Fourth, accuracy-level evaluations hide two problems. One is answer churn at 5–6 bits, which matters when users expect consistent answers. The other is silent, fluent errors at 4 bits, which downstream filters will not catch. Teams deploying INT4 models for code or factual assistance should run task-specific regression suites rather than relying on perplexity. Surprisingly, WIKITEXT-2 perplexity was as good a pooled failure detector as task KL (AUC 0.98), because both explode at 3 bits. Neither caught the subtle INT4 code failures: the perplexity ratio was only 1.12–1.21 at INT4 for models that lost 23–29% on HUMANEVAL.

**How should KL be used?** KL is cheap and label-free, and it reliably flags configurations that are already broken. It does not, however, anticipate a cliff one precision step ahead, because it grows geometrically and crosses any fixed threshold together with accuracy. A single threshold is also unsafe across tasks: code fails at about 0.04 nats, while factual QA is healthy at about 0.4. Task-calibrated thresholds, or the *slope* of KL between adjacent levels, are more promising early-warning signals. We leave both to future work.

### 5.1 Limitations

**Quantizer coverage.** We use weight-only RTN, which is pessimistic, plus two AWQ checkpoints. Running GPTQ, AWQ or other calibrated quantizers at every bit width, or quantizing activations and the KV cache, may shift the cliffs by about a bit. Our fake-quantized weights are stored in bf16, which slightly re-rounds the 8-bit grid; this effect is negligible.

**Model coverage.** We study seven models, and PHI-3-MINI only at bf16, 4 and 3 bits. There are only two LLAMA-3 sizes, one PHI-3-MINI size, and no models above 8B (a 14B bf16 model did not fit on the shared GPUs). Our size conclusions therefore rest heavily on the 0.5B and 7–8B models.

**Statistical resolution.** With 100–400 items per task, per-cell confidence intervals are roughly $\pm$4–8 points, so individual INT4 drops below about 5 points are not resolvable. Critical precisions use point estimates. Greedy decoding gives one deterministic output per configuration, so we report no seed variance, and we did not test sensitivity to prompt format.

**Metrics and data.** The creative-writing judge (Qwen3-8B-AWQ) shares a family with several evaluated models and was not validated against human ratings. Judge scores compress quality differences, and IFEVAL has a floor of about 0.1–0.4 even for gibberish, so the "robust" scores on these tasks partly reflect the metric. The flip analysis shows, however, that IFEVAL answers churn at 5 bits as much as GSM8K answers. GSM8K, TRIVIAQA, SQUAD and HUMANEVAL are public and were likely seen in pre-training. This contamination affects absolute scores more than retention.

**KL measurement.** We measure KL on teacher-forced bf16 responses (first 60 items per task, at most 256 tokens) using a top-128 lower bound. KL along the quantized model's own trajectory could differ. Finally, the cliff definitions (0.85 retained; a 0.15 drop) are pre-registered but arbitrary. The location of the steepest drop does not depend on them.

**Broader impact.** Quantization makes models cheaper and more accessible. Our results show that it can also introduce fluent, confident errors, e.g., wrong facts or subtly broken code, that users cannot easily see. We hope task-resolved evaluation helps deployers avoid shipping such silent regressions.

## 6 Conclusion

We mapped where weight-quantized language models break across six task types, seven models from 0.5B to 8B, and seven precisions. With weight-only group-128 round-to-nearest quantization, accuracy is flat down to 5–6 bits. At 4 bits it erodes in a task-dependent order: code is hit hardest, followed by factual QA and arithmetic, then reading and creative writing, with instruction following least affected. It collapses between 4 and 3 bits for models up to 3.8B and between 3 and 2 bits for 7–8B models. Larger models degrade more gracefully, though not monotonically. KL divergence from the bf16 model marks the collapse with a pooled threshold near 0.2–0.3 nats, but it does not anticipate the collapse, and the right threshold depends strongly on the task. The key takeaway is

that *the INT4 decision should be validated per task, because the failures that matter at 4 bits are fluent, silent, and invisible to perplexity*.

Future work should repeat the sweep with GPTQ and AWQ at 2–4 bits, to test whether the task ranking survives better quantizers, and extend it to 14–70B models. It should also test KL slopes and per-task calibrated thresholds as early-warning signals, check whether QWEN2.5-3B's fragility traces to specific outlier channels, and validate the creative-writing judge against human ratings.

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# A Additional Results

Table 7: bf16 reference scores with 95% bootstrap confidence intervals. Metrics follow table 1. The highest score per task is in **bold**.

| Model | Arith. (GSM8K) | Fact. QA (TRIVIAQA) | Code (HUMANEVAL) | Instr. (IFEVAL) | Reading (SQUAD) | Creative |
|---|---|---|---|---|---|---|
| QWEN2.5-0.5B | 0.460 [0.40,0.52] | 0.140 [0.10,0.17] | 0.329 [0.26,0.40] | 0.260 [0.21,0.31] | 0.681 [0.64,0.72] | 0.458 [0.42,0.50] |
| LLAMA-3-1B | 0.384 [0.32,0.44] | 0.245 [0.20,0.29] | 0.335 [0.27,0.41] | 0.410 [0.36,0.47] | 0.724 [0.69,0.76] | 0.794 [0.77,0.81] |
| QWEN2.5-1.5B | 0.744 [0.69,0.80] | 0.312 [0.27,0.36] | 0.591 [0.52,0.66] | 0.413 [0.36,0.47] | 0.828 [0.80,0.86] | 0.684 [0.65,0.71] |
| QWEN2.5-3B | 0.860 [0.82,0.90] | 0.398 [0.35,0.45] | 0.701 [0.63,0.77] | 0.603 [0.55,0.65] | 0.779 [0.75,0.81] | 0.835 [0.82,0.85] |
| PHI-3-MINI | 0.852 [0.80,0.90] | 0.542 [0.49,0.59] | 0.707 [0.64,0.77] | 0.463 [0.41,0.52] | 0.777 [0.75,0.81] | 0.810 [0.80,0.82] |
| QWEN2.5-7B | **0.900** [0.86,0.94] | 0.545 [0.49,0.59] | **0.829** [0.77,0.88] | **0.700** [0.65,0.75] | **0.867** [0.84,0.89] | **0.856** [0.84,0.87] |
| LLAMA-3-8B | 0.812 [0.76,0.86] | **0.677** [0.63,0.72] | 0.671 [0.60,0.74] | 0.687 [0.63,0.74] | 0.768 [0.74,0.80] | 0.851 [0.84,0.86] |

Table 8: Retained fraction per model and task at 4 bits (RTN group-128), 3 bits, and with the official AWQ INT4 checkpoint (QWEN2.5-3B/7B only). Values below 0.85 are underlined. AWQ lifts every QWEN2.5-3B task that RTN INT4 pushes below 0.85 back above 0.90. Results at 8, 6, 5 and 2 bits are shown in figure 1.

| | Arith. | | | Fact. QA | | | Code | | | Instr. | | | Reading | | | Creative | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | 4b | 3b | AWQ | 4b | 3b | AWQ | 4b | 3b | AWQ | 4b | 3b | AWQ | 4b | 3b | AWQ | 4b | 3b | AWQ |
| QWEN2.5-0.5B | 0.66 | 0.03 | – | 0.79 | 0.05 | – | 0.76 | 0.00 | – | 1.00 | 0.49 | – | 0.87 | 0.09 | – | 0.70 | 0.45 | – |
| LLAMA-3-1B | 0.97 | 0.17 | – | 0.90 | 0.30 | – | 0.91 | 0.15 | – | 1.08 | 0.59 | – | 0.81 | 0.58 | – | 0.98 | 0.67 | – |
| QWEN2.5-1.5B | 0.86 | 0.05 | – | 0.88 | 0.27 | – | 0.77 | 0.01 | – | 0.99 | 0.31 | – | 0.96 | 0.27 | – | 0.97 | 0.31 | – |
| QWEN2.5-3B | 0.90 | 0.00 | 0.95 | 0.83 | 0.00 | 0.94 | 0.71 | 0.00 | 0.91 | 0.90 | 0.19 | 0.94 | 0.83 | 0.00 | 0.93 | 0.87 | 0.14 | 0.98 |
| PHI-3-MINI | 0.96 | 0.65 | – | 0.87 | 0.43 | – | 0.73 | 0.10 | – | 0.90 | 0.78 | – | 1.03 | 0.79 | – | 0.94 | 0.81 | – |
| QWEN2.5-7B | 0.99 | 0.64 | 0.98 | 0.92 | 0.67 | 0.94 | 0.99 | 0.73 | 0.96 | 0.94 | 0.72 | 0.98 | 0.97 | 0.71 | 1.01 | 1.00 | 0.93 | 1.00 |
| LLAMA-3-8B | 0.97 | 0.73 | – | 0.97 | 0.69 | – | 1.01 | 0.62 | – | 0.99 | 0.85 | – | 1.01 | 0.91 | – | 1.01 | 0.97 | – |

Table 9: WIKITEXT-2 perplexity ratio (quantized / bf16). Perplexity barely moves at 4 bits (1.06–1.27×), even for models that lose 23–29% on HUMANEVAL at that precision. QWEN2.5-3B is an outlier at 3 bits (743×).

| Model | 8 bits | 6 bits | 5 bits | 4 bits | 3 bits | 2 bits |
|---|---|---|---|---|---|---|
| QWEN2.5-0.5B | 1.00 | 1.01 | 1.04 | 1.21 | 4.26 | $2.5 \times 10^4$ |
| LLAMA-3-1B | 1.00 | 1.01 | 1.04 | 1.27 | 3.46 | $2.0 \times 10^4$ |
| QWEN2.5-1.5B | 1.00 | 1.01 | 1.03 | 1.12 | 2.65 | $1.3 \times 10^5$ |
| QWEN2.5-3B | 1.00 | 1.01 | 1.02 | 1.13 | 743 | $5.3 \times 10^4$ |
| PHI-3-MINI | – | – | – | 1.13 | 1.92 | – |
| QWEN2.5-7B | 1.00 | 1.00 | 1.02 | 1.06 | 1.73 | $6.9 \times 10^5$ |
| LLAMA-3-8B | 1.00 | 1.00 | 1.01 | 1.08 | 1.77 | $6.8 \times 10^4$ |

## B Implementation Details

**Quantization.** Algorithm 1 summarizes the group-wise round-to-nearest quantizer. When a model had to be split across GPUs with tensor parallelism, we asserted that the input dimension of every shard remained divisible by 128, so that groups matched the unsharded layout. Between precision levels, the original bf16 weights are restored from a CPU copy before the next quantization.

**Algorithm 1** Group-wise asymmetric round-to-nearest fake quantization

**Require:** weight matrix $\boldsymbol{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, bits $b$, group size $g = 128$

1: **for** each row $i$ and each group $G$ of $g$ consecutive input columns **do**

2: $\quad w_{\min} \leftarrow \min_{j \in G} \boldsymbol{W}_{ij}, \quad w_{\max} \leftarrow \max_{j \in G} \boldsymbol{W}_{ij}$

3: $\quad s \leftarrow (w_{\max} - w_{\min})/(2^b - 1), \quad z \leftarrow \text{round}(-w_{\min}/s)$ ▷ *scale and zero point*

4: $\quad q_j \leftarrow \text{clip}(\text{round}(\boldsymbol{W}_{ij}/s) + z,\ 0,\ 2^b - 1)$ for $j \in G$ ▷ *integer code*

5: $\quad \boldsymbol{W}_{ij} \leftarrow s\,(q_j - z)$ for $j \in G$ ▷ *de-quantize back to bf16*

6: **return** $\boldsymbol{W}$

**Generation settings.** Greedy decoding with temperature 0 and seed 0, with each model's chat template. The generation limits are 512 tokens for GSM8K and HUMANEVAL, 32 for TRIVIAQA and SQUAD, 768 for IFEVAL, and 400 for stories. Per-level generation took about 1–2 minutes for 0.5–1.5B models, about 8 minutes for 7–8B models on a 20 GB GPU slice, and 13–33 minutes for PHI-3-MINI and QWEN2.5-3B on a busy shared GPU. Software: vLLM 0.32.0, PyTorch 2.13, transformers 5.19, Python 3.12; statistics with SciPy, statsmodels and scikit-learn.

**KL approximation.** Using the bf16 top-128 log-probabilities plus a lumped tail bucket gives a lower bound on the exact KL. On QWEN2.5-0.5B, the approximation matched the exact KL to within about 1% (e.g., 0.3624 vs. 0.3661 and 0.0929 vs. 0.0932 nats).